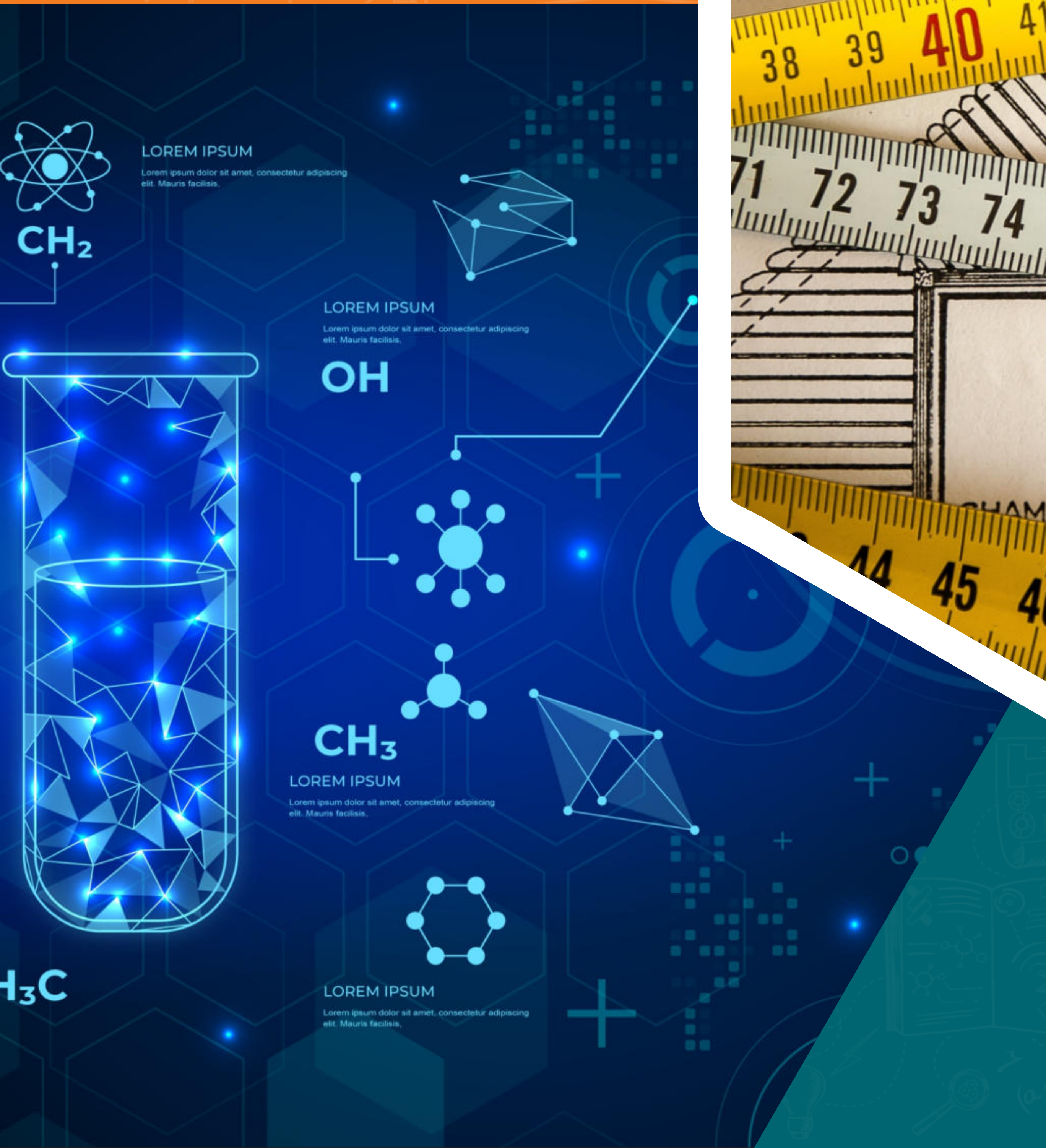
UNIT 10

DC CIRCUITS



OHM'S LAW:

Introduction: In 1826, George Simon Ohm made an investigation of the relation between the potential difference across a conductor and the current flowing through it.

Statement: "The current through a conductor is directly proportional to the potential difference between its ends provided its physical state remains constant."

Mathematically:

$$I \propto \Delta V$$

$$I = K\Delta V$$

Where 'K' is known as the constant of proportionality and is called 'conductance,' which is the inverse of resistance.

$$\text{i.e. } K = \frac{1}{R}$$

$$I = \frac{1}{R}\Delta V$$

$$\Delta V = IR$$

CONDUCTANCE:

"It is an electrical property which depends on the nature of material. It is the facility to the flow of electric current through a conductor."

- Conductance is the opposite of resistance.
- $K = \frac{1}{R}$
- The S.I. unit of conductance is Siemens (S) [Unit named after German scientist Werner von Siemens].
- Both R and K depend upon the shape and size of the conductor.

RESISTANCE:

"Opposition offered by any conductor during the flow of charge is called resistance."

$R = \frac{V}{I}$

- It is represented by 'R.'
- Resistance is measured in V.A-1 = Ω (Ohm). Ω (Ohm).

FACTORS AFFECTING RESISTANCE:

The resistance of a conductor is due to the collision of free electrons with the atoms. It has been found experimentally that at constant temperature, the resistance of a conductor depends upon the following factors:

Length of conductor: Length of conductor is directly proportional to the resistance.

$$R \propto L \quad \text{..............} \quad (1)$$

Cross sectional area of conductor: Area of cross-section of conductor is inversely proportional to resistance.

$$R \propto \frac{1}{A} \quad \text{...........} \quad (2)$$

From equation (1) and (2),

$$R \propto \frac{L}{A}$$

$$R = (\text{Constant}) \times \frac{L}{A}$$

$$R \propto \frac{L}{A}$$

Where 'ρ' is known as resistivity or specific resistance of the material of conductor.

RESISTIVITY:

"The resistivity of a material of a conductor is the resistance of its unit area of cross-section and per unit of its length."

$$\rho = \frac{R \times A}{A}$$

- It is represented by 'ρ'.
- Its unit is Ohm meter (Ω -m).

EFFECT OF TEMPERATURE ON RESISTANCE:

Resistance of conductor is due to the collisions which the free electrons encounter with the atoms or ions of the lattice. As temperature of the conductor rises, the amplitude of the vibration of the ions in the lattice increases.

This increases the probability of collision of free electrons with them. Thus, the lattice atoms offer a bigger target of collisions. The 'drift velocity' thus decreases. This decreases current. Hence, resistance of conductor increases. Experimentally, the change in resistance of a conductor with temperature is nearly linear over a wide range of temperature above and below 0°C.

Let the resistance of a conductor at 0°C = R_0

Let the resistance of a conductor at T°C = R_T

Change in temperature = $\Delta T = T_2 - T_1$

Change in resistance = $\Delta R = R_T - R_0$

By combining above equations,

$$\Delta R \propto R0 \quad \text{..............................} \quad (1)$$

$$\Delta R \propto \Delta T \quad \text{.........................} \quad (2)$$

Where 'α\alphaα' is known as constant of proportionality and is known as temperature 'coefficient of resistance'.

$$RT - R0 = \alpha R0 \Delta T$$

$$RT = R0 + \alpha R0 \Delta T$$

Temperature coefficient of resistance:

"The fractional change in resistivity per unit kelvin is called temperature coefficient of resistance."

- It is denoted by 'α'.
- Its unit is K−1.

$$\alpha = \frac{\Delta R}{R0 \Delta T}$$

Relation between temperature co-efficient and resistivity:

We know that,

$$\alpha = \frac{\Delta R}{R0 \Delta T}$$

$$\alpha = \frac{RT - R0}{R0 \Delta T} \quad \text{..........} \quad (1)$$

Where,

$$R = \frac{L}{A}$$

By putting it in equation (1),

$$\alpha = \frac{L}{A} \times \frac{PT - PO}{PO\frac{L}{A}\ \Delta T}$$

$$\alpha = \frac{L}{A} \times \frac{PT - PO}{PO\frac{L}{A}\ \Delta T}$$

We can write above equation as,

$$R_T = R_0\,(1 + \alpha\Delta T)$$

INTERNAL RESISTANCE:

"Internal resistance refers to the opposition to the flow of current within the cells or batteries or EMF source themselves resulting in the generation of heat."

TERMINAL VOLTAGE:

"When current is drawn from a cell, the potential difference between the electrodes of the cell is called its terminal voltage."

VOLTAGE DROP (ACROSS INTERNAL RESISTANCE):

"As a current flows through the internal resistance of the source, there is a voltage drop which is called voltage drop across internal resistance."

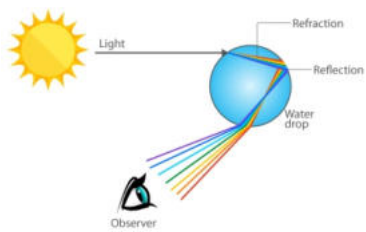
TERMINAL POTENTIAL DIFFERENCE EQUATION:

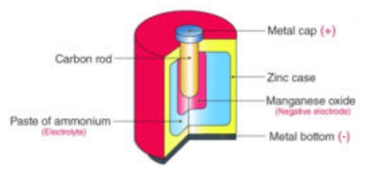
Let us consider a load or an external resistance R is connected across a cell of e.m.f. E. As the cell maintained steady current 'I' in circuit,

Due to the presence of internal resistance 'r' of cell, the total resistance of the circuit would be (R + r). From Ohm's law,

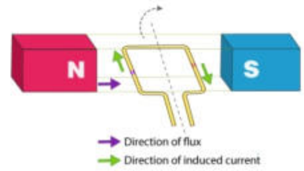
$$VT = E - Ir$$

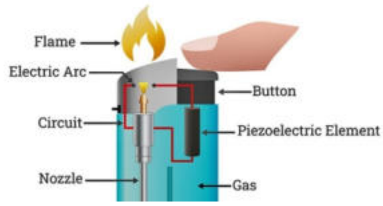
Primary Source of emf	Example
Light	A solar or photovoltaic cells and solar module or panel converts solar light to electric energy. These are made of semiconducting, light-sensitive material which makes electrons available when struck by the light energy.
Chemical Reaction	A battery or voltaic cell directly converts chemical energy into electric energy. It consists of two electrodes in an electrolyte solution. One electrode connects to the (+) or positive terminal (anode), and the other (-) or negative terminal (cathode).





Heat	Thermocouple converts heat energ directly into electric energy. Whe we supplied heat to the hot junctio electrons start moving from or negative charge on one and positive charge on the other.
Mechanical-Magnetic	An electric generator converts mechanical-magnetic energy into electric energy. To produce this, we use a generator such as in an engine, a turbine, or a dynamo.
Piezoelectric Effect	In this type, substance is used which can generate an electric charge when a mechanical force is applied. The materials that are piezoelectric are crystals. An example of piezoelectric is the piezo gas igniter.





POWER DISSIPATION IN RESISTOR:

When P.d. V is applied to the two ends of a conductor, current / flows through it for time 't'. The charge transported is Q = I × t. As the charge moves across the conductor, they lose potential energy, because they collide with the lattice atom of the conductor. The vibrational energy of the lattice atom increases, which appears as heat. The energy dissipated as heat in the conductor is equal to the P.E lost by the charges.

The energy dissipated as heat is E = V × Q.

Since power is the rate of energy, the power dissipated as heat is:

$$P = \frac{E}{t} = \frac{V \times Q}{t}$$

$$P = v\left(\frac{Q}{T}\right)$$

$$P = V I$$

$$\therefore I = \frac{P}{V}$$

According to Ohm's law V= IR:

$$P = (IR)\, I$$

$$P = I^2 R$$

Since $I = \frac{V}{R}$,

$$P = \frac{V2}{R2} \times R$$

$$P = \frac{V2}{R}$$

"Electrical power is the rate at which electrical energy is converted into other forms of energies."

MAXIMUM POWER TRANSFER THEOREM:

"Maximum external power can be obtained from a source with a finite internal resistance, if the resistance of the load is equal to the resistance of the source."

THERMOELECTRICITY:

"Conversion of heat energy into electrical energy with the help of thermocouple is called thermoelectricity."

THERMOCOUPLE:

"A thermocouple is a sensor that measures temperature. It consists of two different types of metals, joined together at one end. When the junction of the two metals is heated or cooled, a voltage is created that can be correlated back to the temperature."

KIRCHOFF'S LAW:

Introduction: In 1847, Gustav Robert Kirchhoff's introduced a pair of laws based on the law of conservation of charge and energy in an electrical circuit. The equivalent impedance of any complex network or circuit can easily be calculated using Kirchhoff's Law.

Classification of Kirchhoff's law: Kirchhoff's laws are classified into two types:

- Kirchhoff's Current Law (KCL)
- Kirchhoff's Voltage Law (KVL)

• Kirchhoff's Current Law (KCL):

Kirchhoff's current law is also known as Kirchhoff's first law or Kirchhoff's junction rule which states that:

"The current that flow into the junction must equal to the current that flows out to the same junction."

$$\sum I_{in} = \sum I_{out}$$

$$\sum I = 0$$

Kirchhoff's first law is based on the principle of law of conservation of charges.

• Kirchhoff's Voltage Law (KVL):

Kirchhoff's voltage law is also known as Kirchhoff's second law or Kirchhoff's loop rule which states that:

"The sum of electromotive forces in a loop is equal to the sum of potential drops into the loop."

$$\sum E = \sum V$$

$$\sum V = 0$$

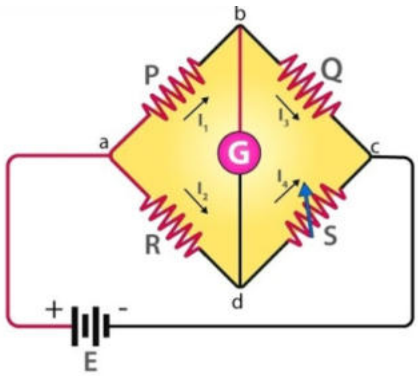
Kirchhoff's second law is based on the principle of law of conservation of energy.

WHEATSTONE BRIDGE:

Introduction: To measure unknown resistance in a circuit, Samuel Hunter Christie invented the Wheatstone bridge in 1833, which Sir Charles Wheatstone later popularised in 1843.

Definition: Wheatstone bridge, also known as the resistance bridge, calculates the unknown resistance by balancing two legs of the bridge circuit. One leg includes the component of unknown resistance.

Working principle: The Wheatstone bridge works on the principle of null deflection, i.e., the ratio of their resistances is equal, and no current flows through the circuit.



Construction: Wheatstone consists on:

- Three fixed resistors.
- One variable resistor.
- Galvanometer.
- EMF source.

Expression for balanced condition (Derivation):

The emf source is attached between points a and c while the galvanometer is connected between points b and d. The current that flows through the galvanometer depends on its potential difference. Current distribution in the circuit is shown in figure.

- Total current by the cell is I, it is distributed to P and R as I_1.
- The current through galvanometer is IG.
- The current in Q is $I_1 - I_G$.
- Through SSS the current is $I - I_1 + I_G$.
- Resistance of galvanometer is G.

By applying Kirchhoff's second law in loop ABDA we get:

$$I_1P + I_G(R - (I_1 - I_G))R = 0 \qquad (1)$$

By applying Kirchhoff's second law in loop BCDB we get:

$$(I_1 - I_1P)(Q - (I_1 - I_G))I_G = 0 \qquad (2)$$

If galvanometer shows no deflection i.e. IG=0I_G = 0IG=0 then, Equation (1) becomes:

$$I_1P\,(I_1 - I_1)\,R = 0$$
$$I_1P + RI = 0$$

By comparing equation (3) and (4):

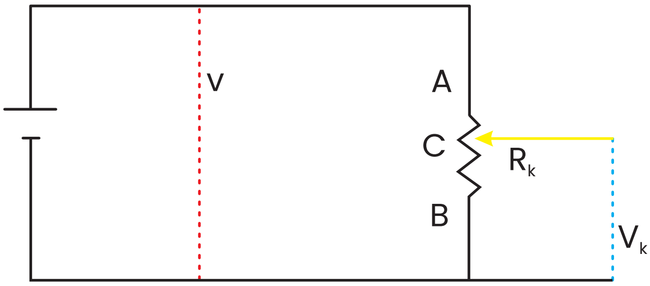
$$RQ + RS = SP + RS$$
$$RQ = SP$$

RHEOSTAT:

"Rheostat is a variable resistor used for controlling the flow of electric current by increasing or decreasing the resistance."

Working of a rheostat as a potential divider:

Let RBC be the resistance of the portion BC of the wire. The current passing through this portion is also I. The PD between the point B and C is given by VBC= IRBC. As RBC increases as we go away from A and decreases as C comes closer to A. This changes the ratio RAC to R and hence a variable voltage can be obtained. If V is regarded as input PD to the potential divider and VBC as the output PD, then VBC can be tapped off and applied to another circuit.



POTENTIOMETER:

"Potentiometer is a three terminal device, consist of a resistance 'R' in the form of a wire on which another terminal can slide, used to measure the EMF of cell."

Principle of working:

Potentiometer working on the principle of that PD across any two points of a wire is directly proportional to the length of a wire.

Working of potentiometer:

Consider a potentiometer of length AB. There are two cells of e.m.f.s E_1 and E_2. Now the positive ends of the cells are connected to point 'A' and the negative ends of the cells are connected to the jockey through galvanometer (G).

When the key is closed and the jockey is moved along wire AB to find the null point (P) where there is no deflection in the galvanometer (G). Let P_1 be the null point when cell E_1 is connected and corresponding length between the end A of the wire to the null point P, be 'L_1'. The potential difference across this length balances emf E_1.

$$E_1 = Kl_1 \qquad \text{(I)}$$

Where, K is the potential gradient of the wire. Then disconnect the cell of e.m.f E_1 and connect the cell of e.m.f E_2 in the circuit. Let P_2 be the null point and let L_2 be the length between the end A of the wire to the null point P_2. Then we have,

$$E_2 = Kl_2 \qquad \text{(ii)}$$

By dividing equation (i) by equation (ii), we get:

$$\frac{E1}{E2} = \frac{L1}{L2}$$

As we measure the value of L1, L2L_1, L_2L1, L2, we can compare the emf of two cells.